

Linear Control Design 1 Formula Collection

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1 Course overview and control concepts

1.1 Linear Control Design 1

1.1.1 PID-based controller structure

$$C_{\text{PID}}(s) = K_P \left(1 + \frac{1}{\tau_i s} + \tau_d s \right)$$

The course focuses on PID-based controller design for linear models. The parameters K_P , τ_i , and τ_d set proportional action, integral action, and derivative action.

1.1.2 Feedback control objective

$$e(t) = r(t) - y_m(t), \quad u(t) = C(s)e(t)$$

The controller acts on the error between the reference and the measured output. Feedback is used to shape tracking, disturbance rejection, stability margins, and robustness.

1.1.3 Control performance keywords

$$\{t_r, t_s, M_p, e_{ss}, \omega_c, \gamma_M, A_M\}$$

The recurring specifications are rise time, settling time, overshoot, steady-state error, crossover frequency, phase margin, and gain margin.

1.2 Course topic order

1.2.1 Inferred lecture structure

1	Introduction and feedback control
2	Block diagrams, control process, hand tuning
3	Transfer functions and Laplace transform
4	Frequency domain and time-domain response
5	Modelling and linearization
6	Bode plots, stability margins, P design
7	Nyquist plot and stability criterion
8	PI-Lead design
9	PI-Lead design specifications
10	Unstable systems
11	Limited systems and P-Lead-Lag design
12	Disturbances, sensitivity, prefilters
13	Feed-forward control

This order follows the numbered slide packages and the exam concepts.

2 Block diagrams and feedback

2.1 Block diagram algebra

2.1.1 Series connection

$$G_{\text{series}}(s) = G_1(s)G_2(s)$$

Series blocks multiply because the output of the first block is the input of the next block.

2.1.2 Parallel connection

$$G_{\text{parallel}}(s) = G_1(s) + G_2(s)$$

Parallel paths with the same input and summed outputs add their transfer functions.

2.1.3 Negative feedback reduction

$$\frac{Y(s)}{R(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}$$

For negative feedback, the loop transfer function $C(s)G(s)H(s)$ appears in the characteristic denominator.

2.1.4 Positive feedback reduction

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

Positive feedback changes the sign in the characteristic denominator and can reduce stability margins.

2.1.5 Unity feedback closed loop

$$T(s) = \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)}, \quad L(s) = C(s)G(s)$$

The loop transfer function $L(s)$ determines both closed-loop stability and frequency-domain robustness.

2.2 Error and sensitivity in feedback

2.2.1 Reference-to-error transfer function

$$\frac{E(s)}{R(s)} = \frac{1}{1 + C(s)G(s)H(s)}$$

Large loop gain reduces tracking error where the feedback model is valid.

2.2.2 Measured feedback signal

$$Y_m(s) = H(s)Y(s)$$

The measurement block represents sensors, filters, encoders, or camera-based measurements used by the controller.

2.3 Hand tuning controllers

2.3.1 P controller

$$u(t) = K_P e(t), \quad C_P(s) = K_P$$

Increasing K_P usually increases speed and reduces static error, but it can reduce damping and robustness.

2.3.2 PI controller

$$u(t) = K_P \left(e(t) + \frac{1}{\tau_i} \int_0^t e(\xi) d\xi \right), \quad C_{PI}(s) = K_P \frac{\tau_i s + 1}{\tau_i s}$$

Integral action increases low-frequency loop gain and can remove step reference error for stable type-0 plants.

2.3.3 PD controller

$$u(t) = K_P \left(e(t) + \tau_d \frac{de(t)}{dt} \right), \quad C_{PD}(s) = K_P (1 + \tau_d s)$$

Derivative action reacts to fast error changes and can add damping, but it amplifies high-frequency noise.

2.3.4 PID controller

$$C_{PID}(s) = K_P \frac{\tau_i \tau_d s^2 + \tau_i s + 1}{\tau_i s}$$

The PID controller combines low-frequency integral action with phase-shaping derivative action.

3 Laplace transform and transfer functions

3.1 Laplace transform

3.1.1 Unilateral Laplace transform

$$X(s) = \mathcal{L}\{x(t)\} = \int_{0^-}^{\infty} x(t)e^{-st} dt$$

The Laplace transform maps differential equations into algebraic equations in s .

3.1.2 Derivative property

$$\mathcal{L}\{\dot{x}(t)\} = sX(s) - x(0^-), \quad \mathcal{L}\{x^{(n)}(t)\} = s^n X(s) - \sum_{k=0}^{n-1} s^{n-1-k} x^{(k)}(0^-)$$

With zero initial conditions, each time derivative corresponds to multiplication by s .

3.1.3 Integral property

$$\mathcal{L}\left\{\int_0^t x(\xi) d\xi\right\} = \frac{X(s)}{s}$$

Integration adds one pole at the origin.

3.1.4 Common Laplace pairs

$x(t)$	$X(s)$
$\delta(t)$	1
$u(t)$	$1/s$
$e^{-at}u(t)$	$1/(s+a)$
$tu(t)$	$1/s^2$
$\sin(\omega t)u(t)$	$\omega/(s^2 + \omega^2)$
$\cos(\omega t)u(t)$	$s/(s^2 + \omega^2)$

These pairs support transfer-function construction and time-response lookup.

3.2 Transfer functions

3.2.1 Transfer function definition

$$G(s) = \left. \frac{Y(s)}{U(s)} \right|_{\text{zero initial conditions}}$$

A transfer function is the input-output model of a linear time-invariant system in the Laplace domain.

3.2.2 Differential equation to transfer function

$$a_n y^{(n)} + \dots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + \dots + b_1 \dot{u} + b_0 u$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \dots + b_1 s + b_0}{a_n s^n + \dots + a_1 s + a_0}$$

The denominator polynomial gives the poles, and the numerator polynomial gives the zeros.

3.2.3 Poles and zeros

$$G(s) = K \frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}$$

Poles p_k define natural modes and stability. Zeros z_i shape phase, transient response, and Bode slope.

3.2.4 Characteristic equation

$$1 + L(s) = 0$$

Closed-loop poles are the roots of the characteristic equation. Stability is determined by these roots, not by the numerator of $T(s)$.

3.2.5 Proper and strictly proper transfer functions

$$\deg N(s) \leq \deg D(s) \quad \text{proper}, \quad \deg N(s) < \deg D(s) \quad \text{strictly proper}$$

Properness is required for physically implementable dynamic systems and feed-forward filters.

4 Frequency and time-domain analysis

4.1 Steady-state gain

4.1.1 DC gain

$$K_{ss} = G(0) = \lim_{s \rightarrow 0} G(s)$$

The steady-state gain is the output/input ratio for constant inputs when the final value exists.

4.1.2 Final value theorem

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

The theorem applies when all poles of $sY(s)$ are in the open left half-plane.

4.2 Frequency response

4.2.1 Frequency response from transfer function

$$G(j\omega) = G(s)|_{s=j\omega}$$

The frequency response gives magnitude and phase for sinusoidal steady-state inputs.

4.2.2 Sinusoidal steady-state output

$$u(t) = A \sin(\omega t) \implies y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$$

The magnitude scales the sinusoid and the phase shifts it.

4.3 First-order response

4.3.1 First-order transfer function

$$G(s) = \frac{K}{\tau s + 1}$$

The parameter K is the DC gain and τ is the time constant.

4.3.2 First-order step response

$$y(t) = K \left(1 - e^{-t/\tau} \right) u(t)$$

At $t = \tau$, the response has reached approximately 63.2% of its final value.

4.4 Second-order response

4.4.1 Standard second-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The natural frequency ω_n sets the time scale, and the damping ratio ζ sets oscillation and overshoot.

4.4.2 Second-order poles

$$p_{1,2} = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

Complex poles occur for $0 < \zeta < 1$ and create oscillatory transients.

4.4.3 Percent overshoot

$$M_p = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}\%$$

Overshoot decreases as damping ratio increases. This formula is used for unit-step second-order specifications.

4.4.4 Settling time approximation

$$t_s \approx \frac{4}{\zeta\omega_n}$$

The approximation corresponds to the common 2% settling-time rule.

4.4.5 Rise time and bandwidth rule

$$t_r \approx \frac{1.8}{\omega_n}$$

This is a compact lookup approximation for underdamped second-order responses.

4.5 Stability from poles

4.5.1 Asymptotic stability

$$\operatorname{Re}(p_k) < 0 \quad \text{for all poles } p_k$$

All poles must lie in the open left half-plane for asymptotic stability.

4.5.2 Unstable poles

$$\exists k : \operatorname{Re}(p_k) > 0$$

Any pole in the right half-plane makes the transfer function unstable.

4.5.3 Marginal stability

$$\operatorname{Re}(p_k) \leq 0, \quad \text{simple poles allowed on } j\omega\text{-axis}$$

Repeated poles on the imaginary axis or at the origin are not marginally stable in the bounded-response sense.

5 Modelling and linearization

5.1 White-box modelling

5.1.1 Newton's second law

$$\sum F = m\ddot{x}$$

Mechanical translation models are obtained by summing forces and applying acceleration relations.

5.1.2 Mass-spring-damper model

$$m\ddot{x} + b\dot{x} + kx = F$$

The mass gives inertia, the damper gives velocity-proportional force, and the spring gives displacement-proportional force.

5.1.3 Mass-spring-damper transfer function

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

With zero initial conditions, the transfer function follows directly from the Laplace transform.

5.1.4 Electrical impedance in Laplace domain

$$Z_R = R, \quad Z_L = sL, \quad Z_C = \frac{1}{sC}$$

RLC circuits can be modelled algebraically in the s -domain using impedances.

5.1.5 Voltage divider in Laplace domain

$$\frac{V_2(s)}{V_{\text{in}}(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

The voltage divider converts circuit topology into a transfer function.

5.2 Nonlinear systems

5.2.1 State-space nonlinear model

$$\dot{x} = f(x, u, t), \quad y = g(x, u, t)$$

Nonlinear models must be linearized before transfer-function and Bode design methods apply.

5.2.2 Equilibrium point

$$0 = f(x_0, u_0), \quad y_0 = g(x_0, u_0)$$

The operating point must make the state derivative zero for constant inputs.

5.2.3 First-order Taylor linearization

$$\Delta \dot{x} = A\Delta x + B\Delta u, \quad \Delta y = C\Delta x + D\Delta u$$
$$A = \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0}, \quad C = \left. \frac{\partial g}{\partial x} \right|_{x_0, u_0}, \quad D = \left. \frac{\partial g}{\partial u} \right|_{x_0, u_0}$$

The variables $\Delta x = x - x_0$, $\Delta u = u - u_0$, and $\Delta y = y - y_0$ describe deviations around the operating point.

5.2.4 State-space to transfer function

$$G(s) = C(sI - A)^{-1}B + D$$

This formula maps a linearized state-space model to an input-output transfer function.

5.3 Black-box modelling

5.3.1 Step-response static gain estimate

$$K_{\text{ss}} \approx \frac{y_{\text{ss}} - y_0}{u_{\text{ss}} - u_0}$$

The static gain can be estimated from steady-state input and output changes.

5.3.2 First-order time constant from data

$$y(t_0 + \tau) = y_0 + 0.632(y_{\text{ss}} - y_0)$$

For a first-order response, the time constant is the time to reach 63.2% of the total step change.

6 Bode plots and stability margins

6.1 Bode magnitude and phase

6.1.1 Magnitude in decibels

$$|G(j\omega)|_{\text{dB}} = 20 \log_{10} |G(j\omega)|$$

Magnitude in decibels converts products of transfer-function factors into sums of asymptotic contributions.

6.1.2 Phase angle

$$\phi(\omega) = \angle G(j\omega)$$

The phase curve is the sum of phase contributions from gain, poles, zeros, integrators, and delays.

6.1.3 Bode form

$$G(s) = K \frac{\prod_i (1 + s/\omega_{z_i})}{s^q \prod_k (1 + s/\omega_{p_k})}$$

Break frequencies ω_{z_i} and ω_{p_k} determine where slopes and phases change.

6.2 Asymptotic Bode rules

6.2.1 Real left-half-plane pole

$$\frac{1}{1 + s/\omega_p} : \quad \Delta M = -20 \text{ dB/dec}, \quad \Delta\phi = -90^\circ$$

A stable real pole lowers the high-frequency magnitude slope and adds negative phase.

6.2.2 Real left-half-plane zero

$$1 + s/\omega_z : \quad \Delta M = +20 \text{ dB/dec}, \quad \Delta\phi = +90^\circ$$

A stable real zero raises the high-frequency magnitude slope and adds positive phase.

6.2.3 Integrator

$$\frac{1}{s} : \quad M_{\text{dB}} = -20 \log_{10} \omega, \quad \phi = -90^\circ$$

An integrator gives infinite DC gain and contributes -20 dB/dec at all frequencies.

6.2.4 Right-half-plane zero

$$1 - \frac{s}{\omega_z} : \quad \Delta M = +20 \text{ dB/dec}, \quad \Delta\phi = -90^\circ$$

A right-half-plane zero has the magnitude effect of a left-half-plane zero but the phase effect of a pole.

6.2.5 Right-half-plane pole

$$\frac{1}{1 - s/\omega_p} : \quad \Delta M = -20 \text{ dB/dec}, \quad \Delta\phi = +90^\circ$$

Right-half-plane poles make the open-loop plant unstable and require Nyquist reasoning for closed-loop stability.

6.3 Second-order Bode factors

6.3.1 Second-order factor

$$G_2(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

At high frequency, a second-order pole pair contributes -40 dB/dec and -180° .

6.3.2 Resonance peak

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}, \quad 0 < \zeta < \frac{1}{\sqrt{2}}$$

A resonance peak indicates low damping and can create multiple crossover frequencies.

6.4 Stability margins

6.4.1 Crossover frequency

$$|L(j\omega_c)| = 1$$

The crossover frequency is where the open-loop magnitude is 0 dB.

6.4.2 Phase margin

$$\gamma_M = 180^\circ + \angle L(j\omega_c)$$

Positive phase margin indicates distance from the critical phase -180° at crossover.

6.4.3 Phase crossover frequency

$$\angle L(j\omega_\pi) = -180^\circ$$

The phase crossover frequency is used to compute gain margin.

6.4.4 Gain margin

$$A_M = \frac{1}{|L(j\omega_\pi)|}, \quad A_{M,\text{dB}} = -20 \log_{10} |L(j\omega_\pi)|$$

The gain margin is the multiplicative gain increase allowed before reaching the critical point.

6.4.5 Bode stability criterion

$$\gamma_M > 0, \quad A_M > 1$$

For stable minimum-phase open-loop systems with a unique crossover, positive phase and gain margins indicate closed-loop stability.

6.4.6 P-controller gain shift

$$L(s) = K_P G(s), \quad 20 \log_{10} |L(j\omega)| = 20 \log_{10} K_P + 20 \log_{10} |G(j\omega)|$$

A positive proportional gain shifts only the magnitude curve and does not change the phase curve.

7 Nyquist plot and stability

7.1 Nyquist curve

7.1.1 Nyquist plot definition

$$\Gamma = \{L(j\omega) : \omega \in (-\infty, \infty)\}$$

The Nyquist plot maps the open-loop frequency response into the complex plane.

7.1.2 Critical point

$$-1 + j0$$

Closed-loop stability is assessed by the relation between the Nyquist curve and the critical point.

7.2 Nyquist stability criterion

7.2.1 Encirclement equation

$$Z = P + N$$

Here P is the number of open-loop right-half-plane poles, N is the net clockwise encirclements of -1 , and Z is the number of closed-loop right-half-plane poles.

7.2.2 Closed-loop stability condition

$$Z = 0$$

The closed loop is stable when the Nyquist encirclements compensate the open-loop unstable poles so that no closed-loop right-half-plane poles remain.

7.2.3 Stable open-loop special case

$$P = 0, \quad N = 0$$

If the open-loop system is stable, the Nyquist curve must not encircle the point $-1 + j0$.

7.3 Margins from Nyquist

7.3.1 Phase margin from unit circle intersection

$$|L(j\omega_c)| = 1, \quad \gamma_M = 180^\circ + \arg L(j\omega_c)$$

On a Nyquist plot, ω_c is where the curve intersects the unit circle.

7.3.2 Gain margin from negative real-axis intersection

$$L(j\omega_\pi) = -a, \quad A_M = \frac{1}{a}$$

The gain margin is read where the Nyquist curve crosses the negative real axis.

7.3.3 P-controller scaling in Nyquist

$$L_K(j\omega) = K_P L(j\omega)$$

Changing K_P radially scales the Nyquist curve. The critical gain occurs when the scaled curve passes through $-1 + j0$.

8 PI-Lead controller design

8.1 PI controller

8.1.1 PI transfer function

$$C_{PI}(s) = K_P \frac{\tau_i s + 1}{\tau_i s}$$

The PI controller adds one zero at $-1/\tau_i$ and one pole at the origin.

8.1.2 PI phase contribution

$$\phi_{PI}(\omega) = \arctan(\omega\tau_i) - 90^\circ$$

The PI part increases low-frequency gain but contributes negative phase near crossover.

8.1.3 PI design ratio

$$N_i = \omega_c \tau_i$$

The ratio N_i places the PI zero relative to the desired crossover frequency.

8.2 Lead controller

8.2.1 Lead transfer function

$$C_{\text{lead}}(s) = \frac{\tau_d s + 1}{\alpha \tau_d s + 1}, \quad 0 < \alpha < 1$$

The lead part adds positive phase and increases high-frequency gain by approximately $1/\alpha$.

8.2.2 Lead maximum phase

$$\phi_{\max} = \sin^{-1} \left(\frac{1 - \alpha}{1 + \alpha} \right)$$

Smaller α gives more phase lead but also more high-frequency amplification.

8.2.3 Lead center frequency

$$\omega_{\max} = \frac{1}{\tau_d \sqrt{\alpha}}$$

The lead part should be centered near the desired crossover frequency.

8.2.4 Lead time constant from crossover

$$\tau_d = \frac{1}{\omega_c \sqrt{\alpha}}$$

This selection places the maximum phase contribution at ω_c .

8.3 PI-Lead controller

8.3.1 PI-Lead transfer function

$$C(s) = K_P \frac{\tau_i s + 1}{\tau_i s} \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

PI-Lead design combines zero steady-state step error with phase-margin improvement.

8.3.2 Crossover gain condition

$$|C(j\omega_c)G(j\omega_c)| = 1$$

The proportional gain is selected after the dynamic PI and lead parts are placed.

8.3.3 Proportional gain from magnitude

$$K_P = \frac{1}{|G(j\omega_c)| |C_{PI,0}(j\omega_c)| |C_{\text{lead}}(j\omega_c)|}$$

Here $C_{PI,0}$ excludes K_P . This formula enforces the desired crossover frequency.

8.3.4 Phase-margin design equation

$$\angle G(j\omega_c) + \phi_{PI}(\omega_c) + \phi_{\text{lead}}(\omega_c) = -180^\circ + \gamma_M$$

The controller phase contributions must lift the open-loop phase to the desired phase margin.

9 Design specifications and tuning

9.1 Time-domain specifications

9.1.1 Steady-state error

$$e_{\text{ss}} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s)$$

Steady-state error is computed with the final value theorem when the closed loop is stable.

9.1.2 Unit-feedback step error

$$e_{\text{ss}} = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}$$

For a unit step reference and unity feedback, this formula links low-frequency loop gain to tracking error.

9.1.3 System type

$$L(s) = \frac{K \prod_i (s - z_i)}{s^q \prod_k (s - p_k)}, \quad q = \text{number of integrators}$$

The system type is the number of poles at the origin in the loop transfer function.

9.1.4 Step, ramp, and parabolic error constants

$$K_p = \lim_{s \rightarrow 0} L(s), \quad K_v = \lim_{s \rightarrow 0} sL(s), \quad K_a = \lim_{s \rightarrow 0} s^2 L(s)$$

These constants determine steady-state error for polynomial references.

9.1.5 Reference error table

Type	step	ramp	parabolic
0	$1/(1 + K_p)$	∞	∞
1	0	$1/K_v$	∞
2	0	0	$1/K_a$

Integral action increases system type and improves low-frequency tracking.

9.2 Frequency-domain specifications

9.2.1 Bandwidth and speed

$$\omega_c \uparrow \implies t_r \downarrow$$

Higher crossover frequency usually gives faster response, limited by phase margin, noise, saturation, and unmodelled dynamics.

9.2.2 Phase margin and damping

$$\gamma_M \uparrow \implies M_p \downarrow$$

Larger phase margin generally means more damping and less overshoot.

9.2.3 P-controller stability limit

$$|K_P G(j\bar{\omega})| = 1, \quad \angle G(j\bar{\omega}) = -180^\circ$$

The critical proportional gain is found at the frequency where the plant phase reaches -180° .

9.3 Ziegler-Nichols and hand tuning

9.3.1 Ultimate gain and period

$$K_u = \text{gain at sustained oscillation}, \quad T_u = \text{oscillation period}$$

The Ziegler-Nichols method uses the stability boundary observed from closed-loop oscillation.

9.3.2 Ziegler-Nichols closed-loop rules

Controller	K_P	τ_i	τ_d
P	$0.5K_u$	—	—
PI	$0.45K_u$	$T_u/1.2$	—
PID	$0.6K_u$	$T_u/2$	$T_u/8$

These rules give heuristic initial controller parameters for subsequent refinement.

9.3.3 Magnitude conversion

$$M_{\text{dB}} = 20 \log_{10} M, \quad M = 10^{M_{\text{dB}}/20}$$

This conversion is used when selecting K_P from a Bode magnitude plot.

10 Unstable systems

10.1 Open-loop unstable plants

10.1.1 Unstable pole condition

$$G(s) \text{ unstable} \iff \exists p_k : \operatorname{Re}(p_k) > 0$$

Open-loop unstable plants cannot be assessed by the stable-plant Bode criterion alone.

10.1.2 Closed-loop stabilization

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}, \quad 1 + C(s)G(s) = 0$$

Controller design must move all closed-loop poles to the open left half-plane.

10.1.3 Nyquist requirement for unstable open loop

$$Z = P + N = 0$$

If the open loop has P right-half-plane poles, the Nyquist encirclement count must give zero unstable closed-loop poles.

10.2 Unstable-system controller design

10.2.1 Stabilizing PI block

$$C_{PI,\text{post}}(s) = K_P \frac{\tau_i s + 1}{\tau_i s}$$

The slides use additional PI action in open loop to stabilize some linearized unstable REGBOT transfer functions.

10.2.2 Lead part for unstable systems

$$C_{\text{lead}}(s) = \frac{\tau_d s + 1}{\alpha \tau_d s + 1}, \quad \tau_d = \frac{1}{\omega_c \sqrt{\alpha}}$$

Lead compensation can increase phase margin or shift crossover when stabilization requires extra phase.

10.2.3 Inner-loop and outer-loop structure

$$L_{\text{inner}}(s) = C_{\text{inner}}(s)G_{\text{inner}}(s), \quad G_{\text{outer}}(s) = \frac{L_{\text{inner}}(s)}{1 + L_{\text{inner}}(s)}$$

Nested controller design stabilizes a fast inner loop before designing the outer loop.

11 Limited systems and Lag control

11.1 Rate and amplitude limitations

11.1.1 Amplitude saturation

$$u_{\text{sat}}(t) = \begin{cases} u_{\text{max}}, & u(t) > u_{\text{max}}, \\ u(t), & u_{\text{min}} \leq u(t) \leq u_{\text{max}}, \\ u_{\text{min}}, & u(t) < u_{\text{min}} \end{cases}$$

Amplitude saturation is nonlinear and can invalidate a linear controller design.

11.1.2 Rate limitation

$$|\dot{u}(t)| \leq R_{\text{max}}$$

Rate limits restrict how quickly the input can change and can create slower or unstable closed-loop behavior.

11.1.3 Integrator windup risk

$$u_I(t) = \frac{K_P}{\tau_i} \int_0^t e(\xi) d\xi$$

When the actuator saturates, the integral state can keep growing although the physical input cannot follow.

11.2 Mitigation strategies

11.2.1 Anti-windup back-calculation

$$\dot{x}_I = e + \frac{1}{T_t} (u_{\text{sat}} - u)$$

Back-calculation reduces integral buildup when saturation separates requested input from applied input.

11.2.2 Reference filtering for rate limits

$$F_r(s) = \frac{1}{\tau_f s + 1}$$

A prefilter can slow reference changes so that the commanded input respects rate limitations.

11.3 P-Lead-Lag control

11.3.1 Lag transfer function

$$C_{\text{lag}}(s) = \frac{\tau_i s + 1}{\frac{\tau_i}{\beta} s + 1}, \quad \beta > 1$$

The lag part increases low-frequency gain relative to high-frequency gain without adding an integrator.

11.3.2 P-Lead-Lag transfer function

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\frac{\tau_i}{\beta} s + 1}$$

P-Lead-Lag design can improve low-frequency accuracy while avoiding unlimited integral buildup.

11.3.3 Lag phase contribution

$$\phi_L(\omega_c) = \arctan \left(\frac{N_i(1 - \beta)}{1 + \beta N_i^2} \right), \quad N_i = \omega_c \tau_i$$

This expression appears in exam and slide design tasks for selecting the lag parameter β .

11.3.4 Non-zero steady-state error with Lag

$$e_{\text{ss}} = \frac{1}{1 + L(0)}$$

A lag controller gives finite low-frequency gain, so zero step error is not guaranteed.

12 Disturbances, sensitivity, and prefilters

12.1 Disturbance transfer functions

12.1.1 Superposition for multiple inputs

$$Y(s) = G_{yr}(s)R(s) + G_{yd_1}(s)D_1(s) + G_{yd_2}(s)D_2(s)$$

Linear systems allow each reference or disturbance contribution to be analysed separately.

12.1.2 Output disturbance after the plant input

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) + \frac{1}{1 + C(s)G(s)}D(s)$$

For an additive output disturbance, feedback suppresses the disturbance by the sensitivity function.

12.1.3 Input disturbance before the plant

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) + \frac{G(s)}{1 + C(s)G(s)}D(s)$$

The plant filters the disturbance before feedback attenuation.

12.1.4 Measurement disturbance

$$Y(s) = \frac{G(s)C(s)}{1 + C(s)G(s)}R(s) - \frac{G(s)C(s)}{1 + C(s)G(s)}N(s)$$

Measurement noise passes through the complementary sensitivity path.

12.2 Sensitivity functions

12.2.1 Sensitivity function

$$S(s) = \frac{1}{1 + L(s)}$$

Sensitivity maps output disturbances and model uncertainty effects in standard unity feedback.

12.2.2 Complementary sensitivity function

$$T(s) = \frac{L(s)}{1 + L(s)}$$

Complementary sensitivity maps reference tracking and measurement-noise transmission.

12.2.3 Input sensitivity

$$G_{ur}(s) = \frac{C(s)}{1 + C(s)G(s)}$$

The control input can become large near bandwidth limits, resonance peaks, or fast reference commands.

12.2.4 Steady-state disturbance error

$$e_{ss,d} = \lim_{s \rightarrow 0} sE_d(s)$$

The final value theorem gives constant-disturbance error when the closed-loop poles are stable.

12.2.5 Disturbance magnitude from Bode plot

$$|y_{ss}| = |G_{yd}(j\omega)| |d_{ss}|$$

At very low frequency, the Bode magnitude approximates constant disturbance sensitivity.

12.3 Prefilters

12.3.1 Reference prefilter

$$Y(s) = \frac{F(s)L(s)}{1 + L(s)}R(s)$$

A prefilter shapes reference tracking without changing the loop transfer function and stability margins.

12.3.2 First-order prefilter

$$F(s) = \frac{1}{\tau_f s + 1}$$

A first-order prefilter reduces fast reference content and can reduce overshoot or input peaks.

12.3.3 Lead prefilter

$$F_{\text{lead}}(s) = \frac{\tau_f s + 1}{\alpha_f \tau_f s + 1}$$

The slides use lead-shaped prefilters to improve reference response without changing feedback robustness.

13 Feed-forward control

13.1 Feed-forward principle

13.1.1 Feed-forward definition

$$u(t) = u_{\text{fb}}(t) + u_{\text{ff}}(t)$$

Feed-forward uses the reference or a measured disturbance directly, without waiting for output error.

13.1.2 Reference feed-forward

$$U(s) = C(s)E(s) + F_r(s)R(s)$$

Reference feed-forward can improve tracking but does not by itself reject unmeasured disturbances.

13.2 Disturbance feed-forward

13.2.1 Disturbance path model

$$Y(s) = G(s)U(s) + D(s)d(s)$$

Here $G(s)$ is the manipulated-input path and $D(s)$ is the disturbance path.

13.2.2 Ideal dynamic disturbance feed-forward

$$F_d(s) = -\frac{D(s)}{G(s)}$$

If $U_{\text{ff}}(s) = F_d(s)d(s)$, the disturbance contribution cancels when the model is exact and $F_d(s)$ is stable and proper.

13.2.3 Filtered feed-forward

$$F_{d,\text{filt}}(s) = -\frac{D(s)}{G(s)}Q(s)$$

The filter $Q(s)$ is added when the ideal ratio is improper, noisy, or too aggressive.

13.2.4 Feed-forward properness condition

$$\deg D(s) - \deg G(s) \leq 0$$

The dynamic feed-forward ratio must be proper for direct implementation. Otherwise a low-pass filter is required.

13.2.5 Disturbance sensitivity with feed-forward

$$G_{yd}(s) = \frac{D(s) + G(s)F_d(s)}{1 + C(s)G(s)}$$

Perfect dynamic feed-forward makes the numerator zero and removes the disturbance path in the nominal model.

13.2.6 Steady-state feed-forward

$$F_d(0) = -\frac{D(0)}{G(0)}$$

Static feed-forward cancels constant disturbances when the DC gains are finite and known.